

AWS Developer Certification

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The certification cover over **40 AWS services**
Creation resources in the cloud cost money.

Creating and AWS account require an email for the root user and and AWS account name. You also need a password to authenticate with your root email.

Cloud computing is the on-demand delivery of compute power, database storage, applications, and other IT resources.
Through a cloud services platform with pay-as-you-go pricing
You can provision exactly the right type and size of computing resources you need

AWS is one of the many cloud provider available on the market!

The five characteristics of cloud computing are the following:

- **On-demand services:** without the need of human interaction
- **Broad network access:** resources can be accessed everywhere
- **Multi-tenancy and resource pooling** multiple customer are served on the same resources
- **Rapid elasticity and scalability** easily create and dispose of resources
- **Measured service** user pay for what they use

AWS has 3 pricing fundamentals:

- **Compute:** Pay for your computational time
- **Storage:** Pay for data stored
- **Data transferred OUT of the cloud**

Shared responsibility Model

You are responsible for the security **IN** the cloud
AWS is responsible for the security **OF** the cloud

AWS global services are: **IAM, CloudFront, WAF and Route53**

1 IAM

AWS IAM is one of the few **global** services

On AWS you apply the **least privilege** principle for the user and services IAM policy

1.1 IAM Security Tools

IAM Credentials Report (account-level) is a service used to lists of all your account's users and the status of their various credentials.

IAM Access Advisor (user-level) is a service used to shows the service permissions granted to a user and when those services were last accessed. You can use this information to revise your policies.

Don't use the root account except for AWS account setup

2 EC2

EC2 User Data script run with **root user**.

EC2 instance are billed on a **per minute (60 sec)** base.

AWS has the following naming convention:

m5.2xlarge

- m: instance class
- 5: generation (AWS improves them over time)
- 2xlarge: size within the instance class

Usage beyond Savings Plans is billed at the On-Demand price.

2.1 Security Groups

Security Groups control how traffic is allowed into or out of our instances.

Security groups only contain **allow** rules.

Security groups rules can reference by IP or by security group.

They control both inbound and outbound traffic Security Groups are linked to the specific **region and VPC** where they are created.

Security group rules are stateful, meaning they allow automatically the return traffic.

If your application is not accessible (time out), then it's a security group issue.

2.2 EBS Volume

They can only be mounted to one instance at a time.

They are bound to a specific availability zone.

It's locked to an Availability Zone, to move a volume you need to take a snapshot.

EBS volume can be attached to a single instance in the same Availability Zone (AZ). whereas EFS file system can be mounted on instances across multiple Availability Zones (AZ)

2.3 EBS Snapshots

Make a backup (snapshot) of your EBS volume at a point in time.

You can move your EBS snapshot to a cheaper archive tier.

You can specify a retention policy (1 day - 1 year) for your deleted EBS snapshot

EBS Snapshot archive are 75% cheaper

2.4 AMI

AMI are customization of an EC2 instance **EC2 Image Builder** is the service use to schedule and organize AMI creation

2.5 EC2 Image Builder

Used to automate the creation of Virtual Machines or container images.

Can be run on schedule.

Spot instances are available at up to a 90% discount compared to the on-demand instance prices

2.6 EC2 Instance store

high performance disk attached to EC2 instances

2.7 EFS

EFS can be mounted to multiple target (100s of EC2 at the same time) and is not available zone bounded.

Amazon EC2 instances can access your file system across AZs, regions, and VPCs, while on-premises servers can access using AWS Direct Connect or AWS VPN.

EFS Infrequent Access (IA) is 92% cheaper compared to EFS Standard.

2.8 AWS FSx

Launch 3th party high performance sys on AWS.

FSx is a windows native shared file system

High availability means running your application / system in at least 2 Availability Zones

3 AWS S3

Maximum size of an s3 object is 50TB.

For uploading object bigger than 5Gb we need to use **multi-part upload**.

S3 bucket policy are used for cross-account bucket/object sharing.

S3 bucket versioning is enabled at bucket level.

Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA) is for data that is accessed less frequently but requires rapid access when needed.

Amazon S3 Glacier Flexible Retrieval (formerly S3 Glacier) is the ideal storage class for archiving data that does not require immediate access but needs the flexibility to retrieve large sets of data at no cost, such as backup or disaster recovery use cases.

3.1 IAM Access Analyzer for S3

Ensure that only the correct people have access to your S3 bucket

3.2 AWS Snowball

On-edge device to **collect and process** data and **migrate** them on AWS

3.3 AWS Storage Gateway

Bridge between the on premise data and AWS data storage option (S3, EBS,...).

Hybrid storage service to allow on-premises to seamlessly use the AWS Cloud. AWS Storage Gateway service provides three different types of gateways – **Tape Gateway, File Gateway, and Volume Gateway** that seamlessly connect on-premises applications to cloud storage, caching data locally for low-latency access.

3.4 S3 Express One Zone

AWS S3 storage fitted for operation that need millisecond retrieval time for stored objects.

Encryption in transit and at rest are a client responsibility for AWS S3

4 Database solutions

4.1 AWS RDS

RDS is a **Relational** Database that use SQL data formatting.

Amazon Aurora RDS engine have the **serverless** option, where we don't provision a capacity for the database but the size is automatically managed by AWS.

When you provision an Amazon RDS Multi-AZ Instance with one standby, Amazon RDS automatically creates a primary DB Instance and synchronously replicates the data to a standby instance in a different Availability Zone (AZ).

Read Replica improves **database scalability**

4.2 ElastiCache

ElastiCache is the managed cache solution for AWS.

AWS takes care of OS maintenance / patching, optimizations, setup, configuration, monitoring, failure recovery and backups

4.3 DynamoDB

The noSQL database service for AWS with serverless database capability.

Fully Managed Highly available with replication across 3 AZ.

Integrated with IAM for security, authorization and administration.

Standard and IA table class.

DAX is the in-memory cache service for dynamoDB used only for the database service.

You can use Amazon DynamoDB to store recommendation results with the **LEAST operational overhead** for any scale.

4.4 Global Tables

Make a DyamoDB database accessible with low latency on multiple region

4.5 Redshift

Amazon Redshift is a fully-managed petabyte-scale cloud-based data warehouse product designed for large scale data set storage and analysis.

OLAP (online analytical process) database with serverless capability.

Redshift can be deployed with serverless capability

4.6 Amazon EMR

Elastic MapReduce DB service that manage hundred of EC2 instances with the possibility to fire "spot instances" used for processing large batch of data streamlined from ML and big data operations.

4.7 Amazon Athena

Serverless database service used to analyze big batch of data with SQL query from object stored on S3

4.8 Amazon QuickSight

Serverless machine learning-powered service use to create interactive dashboard

4.9 DocumentDB

MongoDB inspired service used to store document.

Fully Managed, highly available with replication across 3 AZ with storage that automatically grows in increments of 10GB

4.10 Amazon Neptune

Fully managed **graph** database Highly available across 3 AZ with up to 15 read replicas.

4.11 Amazon Glue

Fully serverless and managed ETL service.

4.12 Amazon Timestream

Fully managed serverless **time series** database

4.13 Amazon Lightsail

Virtual servers, storage, databases, and networking.

Good for people with little cloud experience Amazon Lightsail is an easy-to-use cloud platform that offers you everything needed to build an application or website, plus a cost-effective, monthly plan. Amazon Lightsail offers several preconfigured, one-click-to-launch operating systems, development stacks, and web applications

5 Infrastructure at scale

5.1 SSM

SSM Service help manage the EC2 instance and on-premise server

6 AWS Budget

You can use AWS Budgets to set reservation utilization or coverage targets and receive alerts when your utilization drops below the threshold you define. Under AWS Budget you can create reservation for: **Cost Budget, Reservation Budget and Usage Budget**

7 AWS Trusted Advisor

AWS Trusted Advisor is an online tool that provides you real-time guidance to help you provision your resources following **AWS best practices** on:

- cost optimization
- security
- fault tolerance
- service limits
- performance improvement.

8 CloudTrail

By default, the log files delivered by CloudTrail to your S3 bucket are encrypted using server-side encryption with Amazon S3 managed keys (SSE-S3)

9 AWS Private Link

AWS PrivateLink enables private connectivity between VPCs and supported AWS services without traffic traversing the public internet. It ensures secure communication for applications and services.

10 AWS Site-to-site

AWS Site-to-Site VPN creates a secure connection between your data center or branch office and your AWS cloud resources. **This connection goes over the public internet.**

Virtual private gateway (VGW)/Transit Gateway and Customer Gateway are the components of an AWS Site-to-Site VPN

11 Amazon Transcribe

You can use Amazon Transcribe to add speech-to-text capability to your applications.

12 Amazon Polly

You can use Amazon Polly to turn text into lifelike speech thereby allowing you to create applications that talk.

13 Shared responsibility model

Under the AWS Shared Responsibility Model, customers are responsible for managing their data, **including data encryption.**

14 AWS Support

You should use **AWS Business Support** if you have production workloads on AWS and want 24x7 phone, email and chat access to technical support and architectural guidance in the context of your specific use-cases.

You should use the **AWS Enterprise On-Ramp Support** plan if you have production/business critical workloads.

You should use the **AWS Developer Support** plan if you are testing or doing early development on AWS and want the ability to get **email based technical support** during business hours as well as general architectural guidance as you build and test.

The **AWS Abuse team** can assist you when AWS resources are used to engage in abusive behavior.

The **Acceptable Use Policy** describes prohibited uses of the web services offered by Amazon Web Services

15 AWS WAF

WAF protect against common exploits at **Layer 7**

16 Amazon Shield

AWS Shield Advanced provides expanded DDoS attack protection for web applications running on the following resources:

- Amazon EC2
- Elastic Load Balancing (ELB)
- Amazon CloudFront
- Amazon Route 53
- AWS Global Accelerator.

17 CloudFront

AWS **Edge location** is a site that CloudFront uses to cache copies of the content for faster delivery to users at any location.

AWS CloudFront is a **global service** in scope.

CloudFront Origins are: **S3 Bucket, VPC Origin, Custom Origin.**

17.1 S3 Transfer Acceleration

Increase transfer speed by transferring file to an AWS edge location which will forward the data to the S3 bucket in the target region

17.2 AWS WaveLength

WaveLength Zones are infrastructure deployments embedded within the telecommunications providers' datacenters at the edge of the 5G networks

18 VPC

A gateway endpoint is a gateway that you specify as a target for a route in your route table for traffic destined to a supported AWS service. The following AWS services are supported:

- Amazon Simple Storage Service (Amazon S3)
- Amazon DynamoDB

18.1 Internet Gateway

An Internet Gateway is a horizontally scaled, redundant, and highly available VPC component that allows communication between instances in your VPC and the internet. Therefore, it imposes no availability risks or bandwidth constraints on your network traffic.

NAT Gateways (AWS-managed) and NAT Instances (self-managed) allow your instances in your Private Subnets to access the internet while remaining private.

18.2 VPC Flow Logs

Capture information about IP traffic going into your interfaces.

Captures network information from AWS managed interfaces too: Elastic Load Balancers, ElastiCache, RDS, Aurora, etc...

18.3 VPC Peering

Connect two VPC, privately using AWS' network.

VPC Peering connection is **not transitive**

18.4 AWS PrivateLink

Most secure & scalable way to expose a service to 1000s of VPCs.

Does not require VPC peering, internet gateway, NAT, route tables..., but requires a network load balancer and ENI.

Transit Gateway: Connect thousands of VPC and on-premises networks together

19 AWS Global Accelerator

AWS Global Accelerator is a service that improves the availability and performance of your applications with local or global users. It provides static IP addresses that act as a fixed entry point to your application endpoints in a single or multiple AWS Regions, such as your Application Load Balancers, Network Load Balancers, or Amazon EC2 instances. AWS Global Accelerator uses the AWS global network to optimize the path from your users to your applications, improving the performance of your traffic by as much as 60%.

20 Security

20.1 AWS Shield

Shield standard is a free service activated by default.

Shield Advance provide protection over DDOS attack 24/7 with a monthly fee of 3000\$ and more sophisticated attack.

20.2 AWS Firewall Manager

Manage security rules in all accounts of an AWS Organization.

Rules are applied to new resources as they are created across all and future accounts in your Organization.

Encryption is enabled by default on **CloudTrail log, AWS S3 glacier and Storage Gateway.**

AWS ACM is free of charge for public TLS certificates.

20.3 AWS Artifact

Portal that provides customers with on-demand access to AWS compliance documentation and AWS agreements.

20.4 Amazon GuardDuty

Intelligent Threat discovery to protect your AWS Account that uses Machine Learning algorithms, anomaly detection, 3rd party data.
Can protect against Cryptocurrency attacks.

20.5 AWS Security Hub

Central security tool to manage security across several AWS accounts and automate security checks with an integrated dashboards showing current security and compliance status.

20.6 Amazon Detective

Amazon Detective analyzes, investigates, and quickly identifies the root cause of security issues or suspicious activities.

20.7 IAM Access Analyzer

Find out which resources are shared externally by defining Zone of Trust and notify when a resource is accessed outside a zone of trust

21 Amazon Macie

Amazon Macie is a fully managed data security and data privacy service that uses machine learning and pattern matching to discover and protect your sensitive data in AWS. Macie automatically provides an inventory of Amazon S3 buckets including a list of unencrypted buckets, publicly accessible buckets, and buckets shared with AWS accounts outside those you have defined in AWS Organizations.

22 Machine Learning and AI

22.1 Amazon Kendra

Fully managed document search service powered by Machine Learning that extract answers from within a document

22.2 Amazon Personalize

Fully managed ML-service to build apps with real-time personalized recommendations.

22.3 AWS Control Tower

Easy way to set up and govern a secure and compliant multi-account AWS environment based on best practices

22.4 AWS Resource Access Manager

Share AWS resources that you own with other AWS accounts or with any within your Organization by avoiding resource duplication.

23 Well-architected framework

The AWS Well-Architected Framework provides guidance on building secure, high-performing, resilient, and efficient infrastructure for cloud based

applications. Based on six pillars:

- operational excellence
- security
- reliability
- performance efficiency
- cost optimization
- sustainability

24 Service Health

While the AWS Health - Service Health Dashboard displays the general status of AWS services; the AWS Health - Your Account Health Dashboard gives you a personalized view of the performance and availability of the AWS services underlying your AWS resources.

25 AWS Config

AWS Config is a service that enables you to assess, audit, and evaluate the configurations of your AWS resources. Config **continuously monitors and records** your AWS resource configurations and allows you to automate the evaluation of recorded configurations against desired configurations. Think resource-specific history, audit, and compliance; think Config.

26 Amazon GuardDuty

Amazon GuardDuty is a threat detection service that monitors malicious activity and unauthorized behavior to protect your AWS account. Amazon GuardDuty analyzes billions of events across your AWS accounts from AWS CloudTrail (AWS user and API activity in your accounts), Amazon VPC Flow Logs (network traffic data), and DNS Logs (name query patterns).

27 Amazon Compute Optimizer

AWS Compute Optimizer helps you identify the optimal AWS resource configurations, such as:

- Amazon EC2 instance types
- Amazon EBS volume configurations
- AWS Lambda function memory sizes

28 Other services

28.1 AWS Infrastructure Composer

Visually design and build serverless applications quickly on AWS and allow to deploy AWS infrastructure code without needing to be an expert in AWS.

28.2 AWS Device Farm

Fully-managed service that tests your web and mobile apps against desktop browsers, real mobile devices, and tablets

28.3 Reliability

Ability of a system to recover from infrastructure or service disruptions, dynamically acquire computing resources to meet demand, and mitigate disruptions such as misconfigurations or transient network issues

28.4 AWS Managed Services

Provides infrastructure and application support on AWS.

AMS offers a team of AWS experts who manage and operate your infrastructure for security, reliability, and availability.

28.5 AWS Cost & Usage Report (AWS CUR)

AWS Cost & Usage Report (AWS CUR) contains the most comprehensive set of cost and usage data available. You can receive reports that break down your costs by the hour or month, by product or product resource, or by tags that you define yourself.

AWS updates the report in your bucket once a day in comma-separated value (CSV) format.

Every AWS account provides its own invoice end of the month. You can get separate invoices for development and production environments by setting up separate AWS accounts for each environment.

Business Perspective - The Business Perspective of the AWS CAF ensures that cloud adoption strategies align with the organization's business objectives. It focuses on strategic planning, value realization, and benefits tracking to ensure cloud adoption drives business success.

Governance Perspective - The Governance Perspective focuses on managing risks, compliance, and financial management in the cloud. While important, it does not directly address aligning IT strategy with business objectives.

Operations Perspective - The Operations Perspective emphasizes day-to-day workload operations, monitoring, and performance management. It does not focus on strategy alignment with business goals.

Platform Perspective - The Platform Perspective is centered on designing, implementing, and optimizing cloud services and applications. It does not deal with aligning IT strategies to business objectives.